

EDUCATION

Degree	Institute	CGPA/Percentage	Year
Bachelor of Technology (CSE)	Indira Gandhi Delhi Technical University For Women	9.21	2021-2025
Senior Secondary (CBSE)	Bal Bharati Public School, GRH Marg	97.2%	2021
Secondary (CBSE)	Bal Bharati Public School, GRH Marg	95.4%	2019

EXPERIENCE

- JPMorganChase | Software Engineer I

Consumer & Community Banking (CCB) - Data Technology AI/ML Solutions team

July 2025 - Present

Hyderabad, Telangana, India

– Built and packaged reusable MLOps/LLMOps tooling including a project-generator automation reducing new GenAI service setup time by 85% and published it as an internal PyPI package for platform wide reuse

– Developing agentic AI systems using LangChain and LangGraph for multi-step, tool-calling workflows that generate automated meeting insights and summaries for banking and wealth management stakeholders

– Core contributor to Project Hermes, an LLM-powered meeting-intelligence platform; built Meeting Assist Agents and Digital Interaction Agents performing context synthesis, summarization, and intent-aware personalization using GPT-based LLM orchestration

– Implemented LLM evaluation and guardrails, including LLM-as-a-Judge with Arize Phoenix for hallucination detection, prompt-quality scoring, and response reliability monitoring improving output trustworthiness for production agents

– Contributed to prompt engineering and iterative prompt tuning for agentic use cases, optimizing for factual grounding, relevance, and consistency across LLM providers

– Integrated distributed inference serving patterns (RayServe) and worked with multi-model access across OpenAI GPT and Amazon Bedrock (Claude, Titan)

– Built LLM observability pipelines using Splunk, Dynatrace, and Arize Phoenix for tracing agent execution, monitoring inference latency, and detecting model drift in production

– Provisioned secure API gateways for agent orchestration and managed containerized agent deployment via ECS/EKS (Kubernetes) using Terraform. Conducted performance and load testing (BlazeMeter, JMeter) and implemented automated verification testing within CI/CD pipelines to validate GenAI services pre-release
- JPMorganChase | Software Engineer Program Intern

Consumer & Community Banking (CCB) - Data Technology AI/ML Solutions team

January 2025 - June 2025

Hyderabad, Telangana, India

– Built a production-grade RAG (Retrieval-Augmented Generation) pipeline on AWS EKS, integrating OpenAI and Amazon Bedrock embedding models with OpenSearch Serverless for semantic vector search and enterprise QA

– Developed an end-to-end LLM powered QA pipeline using Lambda for document chunking, embedding generation, and KNN/ANN vector retrieval — with ECS-hosted inference APIs serving internal teams

– Contributed to a CI/CD-enabled GenAI serving framework on EKS with a Chainlit-based conversational UI, supporting multimodular deployment and rapid prototyping of LLM applications

– Evaluated the Smart SDK for agentic LLM workflows, benchmarking hallucination rates and response quality across OpenAI and Bedrock models and prompt strategies. Implemented an LLM-as-a-Judge evaluation framework using Eval Agents and Instajudge, sending traces to Arize Phoenix for real-time hallucination detection and response-quality monitoring

– Built LLM guardrails for prompt-injection resistance, output validation, and content/PII filtering to improve production safety of GenAI applications

– Supported Project Hermes agent and orchestrator infrastructure on ECS, contributing to autoscaling, prompt evaluation for Meeting Assist and Digital Interaction Agent use cases, and UAT/production deployment validation
- The University of Winnipeg | Research Scholar

Mitacs Globalink Research Internship

July 2024 - January 2025

Winnipeg, Manitoba, Canada

– Conducted research under Professor Camilo Valderrama as part of the Mitacs Globalink Research Internship, focusing on enhancing machine learning models through automated feature engineering.

– Designed **Feature-Gen**, a Python library for automated feature engineering using genetic algorithms, boosting model accuracy while ensuring interpretability

– Optimized feature sets via NSGA-II, achieving 54.5% fewer features and 1.5% higher F1 scores across diverse datasets

– Integrated ensemble learning (XGBoost, SVM) and non-linear transformations to enhance machine learning models
- JPMorganChase | Software Engineer Program Intern

Consumer & Community Banking (CCB) - Data Technology Regulatory External Data Services team

June 2024 - July 2024

Hyderabad, Telangana, India

– Collaborated within the REDS team to ensure accurate and compliant data reporting for external regulatory authorities by modernizing legacy systems and automating critical workflows

– Automated data validation checks for regulatory reporting, ensuring compliance with external requirements. Utilized Java Spark for efficient data processing and deployed on AWS EMR for scalable execution

– Migrated a legacy PL/SQL-based reporting system to a Spark-powered ETL pipeline, improving processing efficiency and reducing runtime

– Designed and deployed automated workflows using AWS services and CI/CD pipelines to streamline report generation
- Omdena Edmonton | Data Analyst

Predicting Housing Affordability of Canada’s metropolitan region

June 2022 - July 2022

Edmonton, Alberta, Canada

- Conducted in-depth predictive analysis using Python, data mining, and quantitative analysis to assess the future trajectory of housing affordability
- Developed a comprehensive heat-zone map through data visualization, highlighting regions with worsening housing affordability trends
- Leveraged open-source data and applied advanced predictive analytics to gain a deep understanding of Canada's housing market, while identifying key factors influencing housing affordability, contributing to informed decision-making for addressing this critical issue

PROJECTS

- **Swarm Intelligence** *Python, Scikit Learn, OpenCV, Deep Learning, Flask*
Artificial Intelligence Project **GitHub**
 - Developed algorithm for optimal task distribution and path planning in multi-agent multi-task systems
 - Used K-Means clustering to assign tasks to agents according to proximity and threading for synchronous movement
 - Used fast solutions to the Travelling Salesman Problem to decide the order of traversal for multi-objective tasks
 - Implemented A* Path Finding algorithm from scratch to find optimal path between two objectives
- **Dominant Colour Detection** *Python, Scikit Learn, NumPy, OpenCV, Matplotlib, Pandas, Seaborn*
Project Jupyter Notebook **GitHub**
 - Implemented K-means clustering for dominant color extraction in images, and optimized convergence speed
 - Performed image segmentation, and analyzed image content for efficient color extraction for image processing
- **Voice Notes** *HTML, CSS, JavaScript, Web Speech API, SpeechRecognition API*
Voice Notes Web App **GitHub**
 - Web App to create and edit new notes by voice command or by typing
 - It allows users to create, edit, save, listen to, and delete notes, providing a comprehensive note-taking experience.
- **The Weather App** *HTML, CSS, JavaScript, OpenWeather API*
Weather App Website **GitHub**
 - Developed a dynamic Weather Web App utilizing the OpenWeather API, offering real-time weather data, dynamic backgrounds, and user-friendly alerts for accurate city input

ACHIEVEMENTS

- **AWS Certified Generative AI Developer – Professional:** Issued by Amazon Web Services (AWS)
- **AWS Certified Developer - Associate and AWS Certified Cloud Practitioner:** Issued by Amazon Web Services (AWS)
- **Software Engineering Innovation Award:** Issued by JPMorganChase in Dec 2025 for contributions to Project Hermes
- **JPMorganChase 10th Global Hackathon:** Secured 8th place globally (431 submissions, 2500+ participants)
- **MITACS Globalink Research Internship:** One of 300 students out of 100,000 applicants worldwide to receive research grant
- **Microsoft MVP Award 2024:** Recognized technical leadership in the Windows Development
- **JP Morgan Code for Good 2023:** Addressed real-world challenges for social impact in a 24-hour Hackathon
- **Uber She++ 2023:** Selected among top 75 girls across the nation from the 19 best engineering colleges in the country
- **Gold Microsoft Learn Student Ambassador:** Empowered peers with tech expertise through Workshops and Tech-Talks
- **Ideate 'N' Solve (Devcation 2022 Hackathon):** Jury Award by Google Developer Student Club
- **Google HASHCODE 2022:** Global rank 1133, All India Rank 241
- **IIT JEE MAINS 2021:** Secured top 2 Percentile among 1 million candidates

PUBLICATIONS

- **Feature Engineering through Two-Level Genetic Algorithm** *Elsevier Journal Link*
Machine Learning with Applications (MLWA) · Jul 4, 2025
- **A Hierarchical Feature Engineering Framework for Enhancing Interpretable Machine Learning Models** *IEEE Link*
IEEE International Conference on Big Data Computing Service and Machine Learning Applications, Sep 2025
- **Comprehensive Evaluation of Anomaly Detection Methods for Time Series** *Springer Nature Link*
International Conference on Data Analytics & Management (ICDAM-2025)

SKILLS

- **Programming/Tools:** Python, C/C++, SQL, HTML, CSS, JavaScript, LaTeX, Git, GitHub
- **Courses:** Software Engineering, Database Management Systems, Object Oriented Programming, Data Structures and Algorithms, Operating System, Operation Management, Big Data, Artificial Intelligence, Computer Vision, Machine Learning, Data Mining, Optimization Algorithms, Natural Language Processing, Predictive Modeling, Data Ethics
- **Soft Skills:** Problem Solving, Analytical thinking, Interpersonal skills, Research, Project Management, Strategist, Teamwork, Collaboration, Avid learner, Proficient writer, Organized and Detail oriented, Adaptability and continuous learning

LANGUAGES

- **English:** Native or Bilingual Proficiency
- **French:** Professional working proficiency
- **German:** Professional working proficiency
- **Hindi:** Native or Bilingual Proficiency